

HITENSH KHARVA

Software & Data Engineer

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SUMMARY

Software and data engineer with 7+ years across AWS backend services, large-scale data platforms, and production AI. At Amazon, builds LLM agents that act autonomously on production infrastructure, architects governed data platforms for AI planning and cross-border logistics, and shipped platforms adopted by 13+ teams and a public AWS CodeArtifact feature.

SKILLS

AI/ML: Amazon Bedrock, Bedrock AgentCore, Strands Agents, MCP, RAG, vector search, LLM agents

Languages and Cloud: Python, SQL, TypeScript, Java, AWS, infrastructure as code (CDK)

Data: Spark (PySpark, Spark SQL), Airflow, Redshift, Iceberg, Glue, Athena, dbt, Snowflake, data modeling

Engineering: system design, distributed systems, observability, data governance, CI/CD, mentoring

EXPERIENCE

Amazon

Seattle, WA

Data Engineer II, Global Logistics

Jun 2026 – Present

- Diagnosed recurring Redshift congestion (up to 260 queued queries) to 17K+ daily BI refreshes on one shared account, then engineered a Python LLM agent on CloudWatch, EventBridge, Step Functions, Lambda, and Bedrock AgentCore that cancels them under a least-privilege guardrail that cannot touch ETL, resolving its first live incident unattended with **zero failures**.
- Architected the data platform for an AI capacity-planning agent from scratch using TypeScript CDK, Spark SQL, S3, Glue, and Lake Formation, replacing hand-refreshed CSVs with **25** weekly-refreshed datasets across **8** logistics domains served through MCP, supporting a goal of cutting planning cycles from 3 months to 1 day.
- Leading the production rollout of a config-driven carrier API service that replaces web scraping, ingesting DCSA vessel schedules through Airflow (MWAA), Lambda, and Spark with idempotent merges into Iceberg, so onboarding a carrier requires one configuration entry.
- Led data engineering for a newly launched cross-border logistics business, gathering EU and US requirements and designing the data model and pipelines that unify 8+ source systems into one source of truth, built as Airflow (MWAA) DAGs of modular Spark jobs with dependency sensors, task-level restarts, and SLA alerting.
- Co-built the team's production multi-agent platform on Bedrock AgentCore and Strands, delivering its Python Lambda deployment pipeline, the autonomous congestion workflow, and the design to migrate the triage assistant onto it, and root-causing a shared-runtime cold-start failure affecting **60%** of invocations.

Data Engineer I, Global Logistics

Jul 2024 – Jun 2026

- Designed and launched a multi-tenant S3-to-data-lake ingestion platform using TypeScript CDK, Lambda, and Spark that replaced hand-built pipelines and was self-onboarded by **13+** teams, and automated **5,000+** access requests for **500+** tables.
- Developed a Python RAG assistant on Lambda, Bedrock (Claude), and OpenSearch Serverless that posts ranked runbook fixes to new tickets in under 15 seconds, reducing on-call analysis time by **80%** and becoming the team's standard triage process.
- Automated ocean-carrier schedule collection with browser-automation agents feeding an S3-to-data-lake pipeline, replacing manual work; a live demo to senior leadership launched the carrier API partnerships that the API service now delivers.
- Resolved a production Redshift performance crisis, added proactive alarms and an hourly workload monitor, moved log retention to S3 and Redshift Spectrum, and rebuilt query-ownership attribution from audit logs (accuracy from **14%** to **98%**).

Amazon Web Services

Seattle, WA

Software Development Engineer, AWS CodeArtifact

Dec 2023 – Jul 2024

- Implemented gem upload and download in the Java and Guice adapter service on ECS for Ruby support in AWS CodeArtifact, covering gem push and install, Bundler requests, .gem payload parsing, and a bounded dependency API, launched to customers in April 2024.
- Developed the Ruby protocol client library from scratch, including asset retrieval and upload, version and package metadata, authentication, and **7** typed error classes, enabling end-to-end service testing before launch.
- Wrote TestNG functional, negative, billing, CloudTrail audit, and canary integration tests that gated the deployment pipeline, and delivered console onboarding, queue alarms, dashboards, and beta-customer and penetration-test access.

Wind River Systems

San Diego, CA

Data Developer

Jun 2023 – Dec 2023

- Developed Fivetran pipelines bringing SurveyMonkey and Hootsuite data into Snowflake, with staging, intermediate, and mart dbt models powering Looker dashboards for the People Branding team.

- Integrated dbt-checkpoint pre-commit hooks into CI/CD, reducing data model review time by **25%**, improving documentation accuracy by **30%**, and raising naming-convention compliance above **90%**.
- Engineered a Python CI pipeline that detects which dbt changes affect which Looker tiles and re-runs only those tiles, and added Spectacles validation for LookML models.

Data Developer Intern

May 2022 – Mar 2023

- Designed Python models that scan, tag, and mask PII fields across Snowflake schemas for data governance.
- Added autoscaling and YAML linting to the dbt CI pipeline, reducing development time by **40%**, and built a Python and SQL tool that compares development and production data.

Bloomstack Technology

Mumbai, India

Backend Python Engineer

Nov 2020 – May 2021

- Engineered a Python and SQL pipeline that ingests **200+** semi-structured data points per day from an external REST API into MySQL and PostgreSQL, enabling consistent analysis and reporting.
- Developed backend modules in Python on the Frappe and ERPNext platform for the company's core product.

Larsen & Toubro Infotech

Pune, India

Software Engineer

Jul 2018 – Oct 2020

- Delivered web and mobile banking products for clients across **9** countries through the full Agile software development life cycle, meeting all project requirements.
- Proposed and implemented automated testing and reporting in Python across **30+** functional areas, improving efficiency **20%**.
- Analyzed and tested database objects and SQL statements with unit and integration tests before production releases, using Python, Django, MySQL, and MongoDB.

EDUCATION AND CERTIFICATIONS

Master of Science in Computer Science, San Diego State University

May 2023

Bachelor of Engineering in Electronics and Telecommunication, University of Mumbai

May 2018

Microsoft Azure Data Engineer Associate | Stanford Algorithms Specialization | IBM Data Engineering Certificate